

CHAPTER 18

interest rate derivatives



The aim of this Chapter...

... is to examine some of the more important fixed-income products and to explain various ways in which to approach their pricing. You will see how it is common market practice to use simple Black–Scholes pricing formulæ in novel ways.

In this Chapter...

- common fixed-income contracts such as bond options, caps and floors
- how to price interest rate products in the consistent partial differential equation framework
- how to price contracts the market way
- path dependency in interest rate products, such as the index amortizing rate swap

18.1 INTRODUCTION

So far in this book I derived a theory for pricing and hedging many different types of options on equities, currencies and commodities. In Chapter 16 I presented the theory for zero-coupon bonds, boldly saying that the model may be applied to other contracts.

In the equity options world we have seen different degrees of complexity. The simple contracts have no path dependency. These include the vanilla calls and puts and contracts having different final conditions such as binaries or straddles. At the next stage of complexity we find the path-dependent contracts such as American options or barriers for which, technically speaking, the path taken by the underlying is important. Many of these ideas are mirrored in the theory of interest rate derivatives.

In this chapter we delve deeper into the subject of fixed-income contracts by considering interest rate derivatives such as bond options, caps and floors, swaptions, captions and floortions, and more complicated and path-dependent contracts such as the index amortizing rate swap.



Time Out...

Pricing methodologies

I'm going to give you some insight into the two main methods for pricing interest rate derivatives. One way is consistent across all instruments but not necessarily accurate, the other is the exact opposite.

The former method is to price contracts using the same stochastic differential equation model for the spot rate and the resulting partial differential equation. Different contracts have different final and boundary conditions. The problem with this approach is that the basic model may not be that accurate. It can therefore be highly dangerous to use this method for pricing popular, highly liquid contracts.

The other approach is to bend and squeeze the instrument so as to make the 'underlying' (suitably defined) look like a lognormal asset. From then on, just apply the basic Black–Scholes formulæ. The main positive point about this rather artificial method is that it is common practice in the market.

18.2 CALLABLE BONDS

As a gentle introduction to more complex fixed-income products, consider the **callable bond**. This is a simple coupon-bearing bond, but one that the issuer may call back on specified dates for a specified amount. The amount for which it may be called back may

be time dependent. This feature reduces the value of the bond; if rates are low, so that the bond value is high, the issuer will call the bond back.

The callable bond satisfies

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial r^2} + (u - \lambda\sigma) \frac{\partial V}{\partial r} - rV = 0,$$

with

$$V(r, T) = 1,$$

and

$$V(r, t_c^-) = V(r, t_c^+) + K_c,$$

across coupon dates. If the bond can be called back for an amount $C(t)$ then we have the constraint on the bond's value

$$V(r, t) \leq C(t),$$

together with continuity of $\partial V / \partial r$.

18.3 BOND OPTIONS

The stochastic model for the spot rate presented in Chapter 16 allows us to value contingent claims such as bond options. A **bond option** is identical to an equity option except that the underlying asset is a bond. Both European and American versions exist.

As a simple example, we derive the differential equation satisfied by a call option, with exercise price E and expiry date T , on a zero-coupon bond with maturity date $T_B \geq T$. Before finding the value of the option to buy a bond we must find the value of the bond itself.

Let us write $Z(r, t; T_B)$ for the value of the bond. Thus, Z satisfies

$$\frac{\partial Z}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 Z}{\partial r^2} + (u - \lambda\sigma) \frac{\partial Z}{\partial r} - rZ = 0 \quad (18.1)$$

with

$$Z(r, T_B; T_B) = 1$$

and suitable boundary conditions. Now write $V(r, t)$ for the value of the call option on this bond. Since V also depends on the random variable r , it too must satisfy Equation (18.1). The only difference is that the final value for the option is

$$V(r, T) = \max(Z(r, t; T_B) - E, 0).$$

This payoff is shown in Figure 18.1.

Figure 18.2 shows the Bloomberg option calculator for bond options. In this case the model used is Black, Derman & Toy.

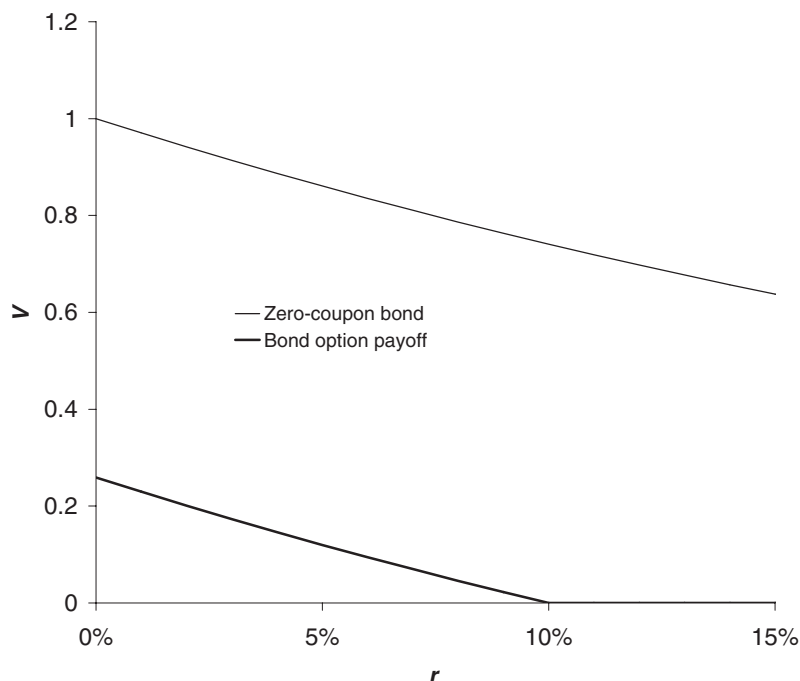


Figure 18.1 Zero-coupon bond price as a function of spot, and the payoff for a call option on the bond.

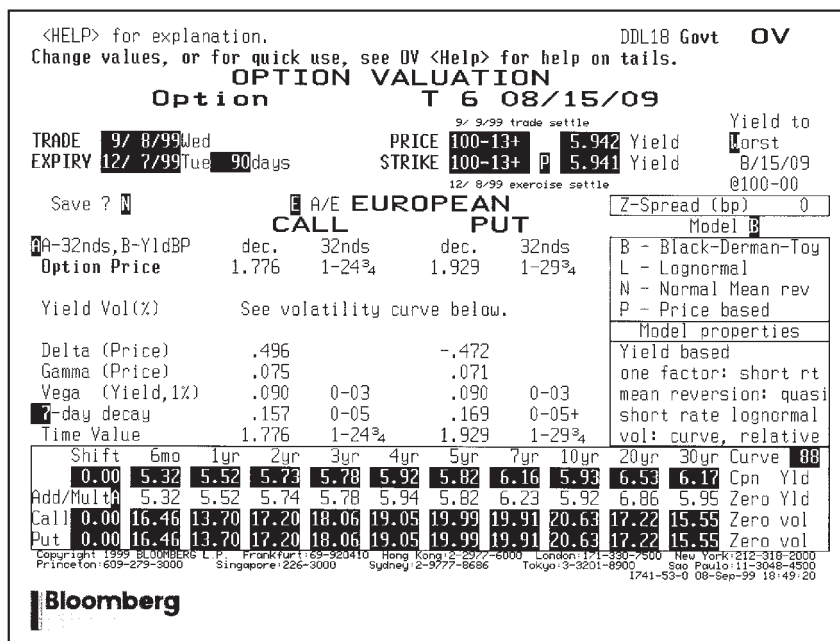


Figure 18.2 Bond option valuation. Source: Bloomberg L.P.

*Time Out...***Must we really solve the partial differential equation?**

No. There are lots of other things you can do. Here's a brief summary of your choices.

- If there's no 'optionality' in the contract, and all cashflows are fixed or floating as in swaps, you should price by discounting using the yield curve. But first convert floating cashflows to fixed as described in Chapter 15.
- Use the above pde approach if you feel comfortable with such concepts, and are happy to solve numerically by finite-difference methods. I hope you will feel happy with this by the end of the book.
- You can use trees, as hinted at in the previous chapter. Easy to understand, but not very sophisticated from a numerical point of view.
- Risk-neutral expectations are always there for you to fall back on. All the details are covered in Chapter 29.
- Finally, perhaps most popular and robust, fudge. This means pretend that you've got an equity derivative and not a fixed-income derivative and use a Black–Scholes-type formula. Some examples are given below.

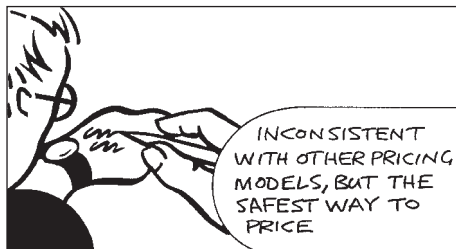
**18.3.1** Market practice

The above is all well and good, but suffers from the problem that any inaccuracy in the model is magnified by the process of solving once for the bond and then again for the bond option. This makes the contract second order, see Chapter 11. When the time comes to exercise the option the amount you receive will, for a call, be the difference between the *actual* bond price and the exercise price, not the difference between the *theoretical* bond price and the exercise price. So the model had better be correct for the bond price. Of course, this model can never be correct, and so we must treat the pricing of bond options with care. Practitioners tend to use an approach that is internally inconsistent but which is less likely to be very wrong. They use the Black–Scholes equity option pricing equation and formulæ assuming that the underlying is the bond. That is, they assume that the bond behaves like a lognormal asset. This requires them to estimate a volatility for the bond, either measured statistically or implied from another contract, and an interest rate for the lifetime of the bond option. This will be a good model provided that the expiry of the bond option is much shorter than the maturity of the underlying bond. Over short time periods, well away from maturity, the bond does behave stochastically, with a measurable volatility.

This method is called 'Black 76' after the famous paper by that author.

The price of a European bond call option in this model is

$$e^{-r(T-t)} (FN(d'_1) - EN(d'_2)),$$



and the put has value

$$e^{-r(T-t)} (EN(-d'_2) - FN(-d'_1)),$$

where F is the forward price of the bond at expiry of the option and

$$d'_1 = \frac{\log(F/E) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}} \quad \text{and} \quad d'_2 = d'_1 - \sigma\sqrt{T-t}.$$

This model should not be used when the life of the option is comparable to the maturity of the bond, because then there is an appreciable **pull to par**, that is, the value of the bond at maturity is the principal plus last coupon; the bond cannot behave lognormally close to maturity because we know where it must end up, this contrasts greatly with the behavior of an equity for which there is no date in the future on which we know its value for certain. This pull to par is shown in Figure 18.3.

Another approach used in practice is to model the yield to maturity of the underlying bond. The usual assumption is that this yield follows a lognormal random walk. By modeling the yield and then calculating the bond price based on this yield, we get a bond that behaves well close to its maturity; the pull to par is incorporated.

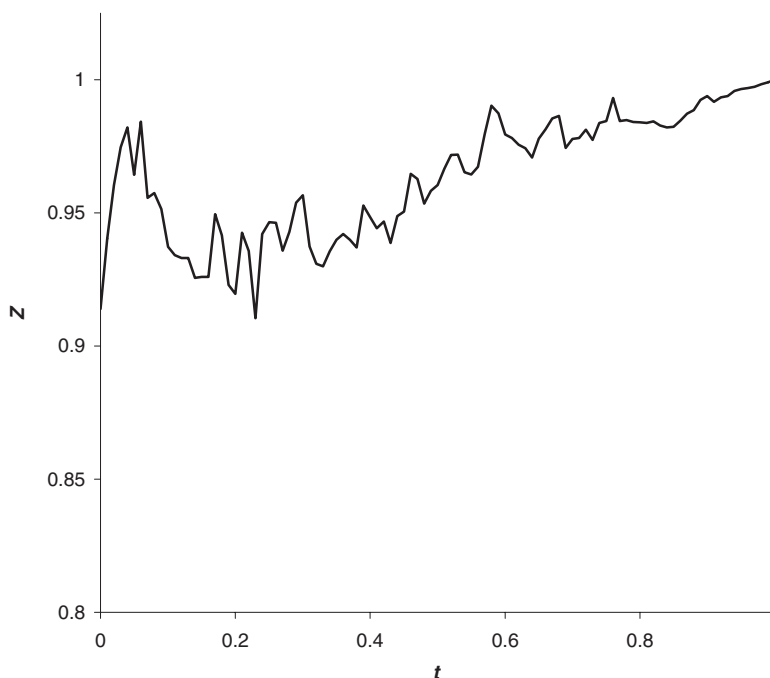


Figure 18.3 The pull to par for a zero-coupon bond.

There is one technical point about the definition of the bond option concerning the meaning of 'price.' One must be careful to use whichever of the clean or dirty price is correct for the option in question. This amounts to knowing whether or not accrued interest should be included in the payoff, see Chapter 14.

18.4 CAPS AND FLOORS

A **cap** is a contract that guarantees to its holder that otherwise floating rates will not exceed a specified amount; the variable rate is thus capped.

A typical cap contract involves the payment at times t_i , each quarter, say, of a variable interest on a principal with the cashflow taking the form

$$\max(r_L - r_c, 0),$$

multiplied by the principal. Here r_L is the basic floating rate, for example three-month LIBOR if the payments are made quarterly, and r_c is the fixed cap rate. These payments continue for the lifetime of the cap. The rate r_L to be paid at time t_i is set at time t_{i-1} . Each of the individual cashflows is called a **caplet**; a cap is thus the sum of many caplets.

The cashflow of a caplet is shown in Figure 18.4.

If we assume that the actual floating rate is the spot rate, i.e. $r_L \approx r$ (and this approximation may not be important), then a single caplet may be priced by solving

$$\frac{\partial V}{\partial t} + \frac{1}{2}w^2 \frac{\partial^2 V}{\partial r^2} + (u - \lambda w) \frac{\partial V}{\partial r} - rV = 0, \quad (18.2)$$

with

$$V(r, T) = \max(r - r_c, 0).$$

Mathematically, this is similar to a call option on the floating rate r .

Figure 18.5 shows the Bloomberg calculator for caps.

A **floor** is similar to a cap except that the floor ensures that the interest rate is bounded below by r_f . A floor is made up of a sum of floorlets, each of which has a cashflow of

$$\max(r_f - r_L, 0).$$

We can approximate r_L by r again, in which case the floorlet satisfies the bond pricing equation but with

$$V(r, T) = \max(r_f - r, 0).$$

A floorlet is thus a put on the spot rate.

18.4.1 Cap/floor parity

A portfolio of a long caplet and a short floorlet (with $r_c = r_f$) has the cashflow

$$\max(r_L - r_c, 0) - \max(r_c - r_L, 0) = r_L - r_c.$$

This is the same cashflow as one payment of a swap. Thus there is the model-independent no-arbitrage relationship

$$\text{cap} = \text{floor} + \text{swap}.$$

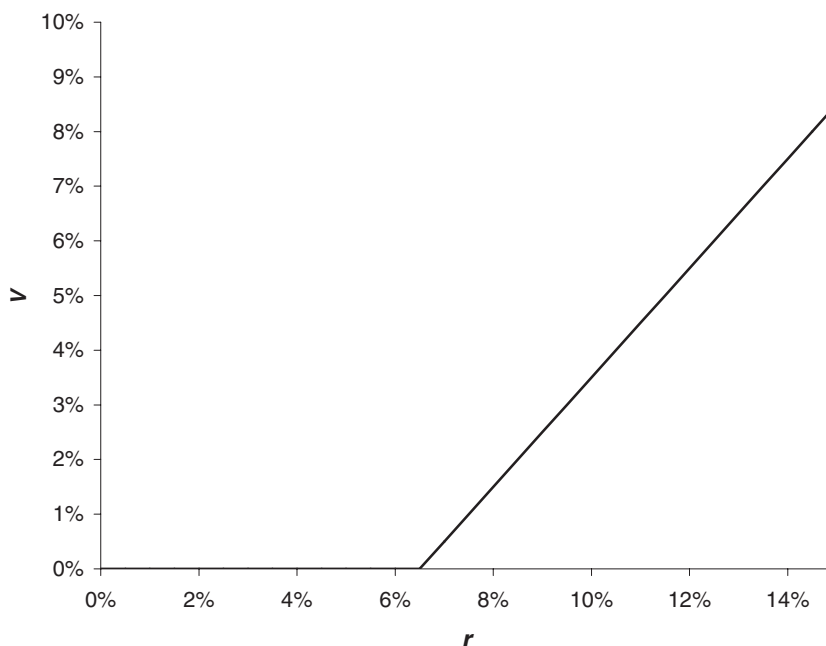


Figure 18.4 The dependence of the payment of a caplet on LIBOR.

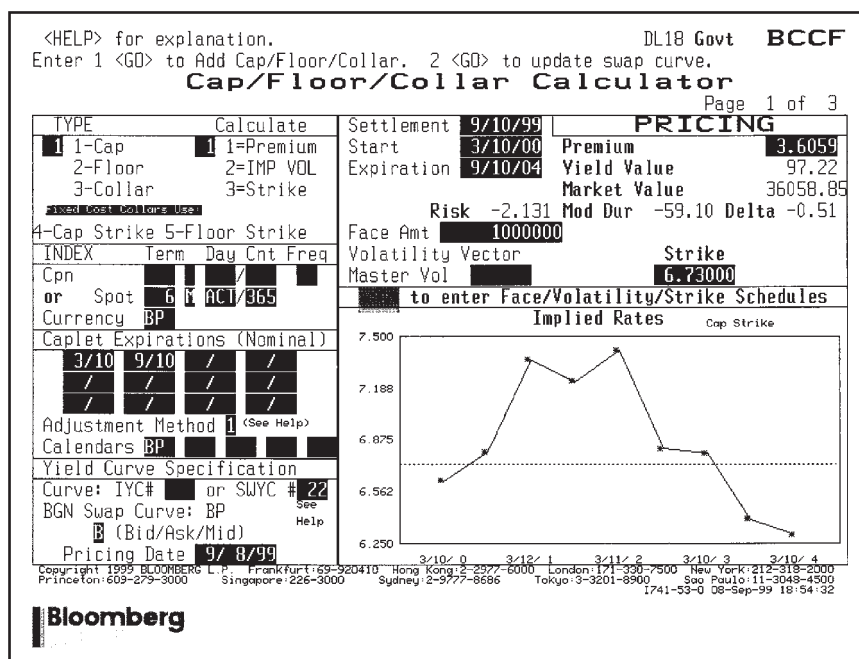


Figure 18.5 Cap/floor/collar calculator. Source: Bloomberg L.P.

18.4.2 The relationship between a caplet and a bond option

A caplet has the following cashflow

$$\max(r_L - r_c, 0).$$

This is received at time t_i but the rate r_L is set at t_{i-1} . This cashflow is *exactly* the same as the cashflow

$$\frac{1}{1 + r_L} \max(r_L - r_c, 0)$$

received at time t_{i-1} , after all, that is the definition of r_L . We can rewrite this cashflow as

$$\max\left(1 - \frac{1 + r_c}{1 + r_L}, 0\right).$$

But

$$\frac{1 + r_c}{1 + r_L}$$

is the price at time t_{i-1} of a bond paying $1 + r_c$ at time t_i . We can conclude that a caplet is equivalent to a put option expiring at time t_{i-1} on a bond maturing at time t_i .

18.4.3 Market practice

Again, because the Black–Scholes formulæ are so simple to use, it is common to use them to price caps and floors. This is done as follows. Each individual caplet (or floorlet) is priced as a call (or put) on a lognormally distributed interest rate. The inputs for this model are the volatility of the interest rate, the strike price r_c (or r_f), the time to the cashflow $t_i - t$, and *two* interest rates. One interest rate takes the place of the stock price and will be the current forward rate applying between times t_{i-1} and t_i . The other interest rate, used for discounting to the present is the yield on a bond maturing at time t_i . For a caplet the relevant formula is

$$e^{-r^*(t_i-t)} (F(t, t_{i-1}, t_i)N(d'_1) - r_c N(d'_2)).$$

Here $F(t, t_{i-1}, t_i)$ is the forward rate today between t_{i-1} and t_i , r^* is the yield to maturity for a maturity of $t_i - t$,

$$d'_1 = \frac{\log(F/r_c) + \frac{1}{2}\sigma^2 t_{i-1}}{\sigma\sqrt{t_{i-1}}} \quad \text{and} \quad d'_2 = d'_1 - \sigma\sqrt{t_{i-1}}.$$

σ is the volatility of the $(t_i - t_{i-1})$ interest rate. The floorlet value is

$$e^{-r^*(t_i-t)} (-F(t, t_{i-1}, t_i)N(-d'_1) + r_c N(-d'_2)).$$

18.4.4 Collars

A **collar** places both an upper and a lower bound on the interest payments. It can be valued as a long cap and a short floor.

18.4.5 Step-up swaps, caps and floors

Step-up swaps, etc. have swap (cap, etc.) rates that vary with time in a prescribed manner.

18.5 RANGE NOTES

The **range note** pays interest on a notional principal for every day that an interest rate lies between prescribed lower and upper bounds. Let us assume that the relevant interest rate is our spot rate r . In this case we must solve

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 \frac{\partial^2 V}{\partial r^2} + (u - \lambda\sigma) \frac{\partial V}{\partial r} - rV + \mathcal{I}(r) = 0,$$

with

$$V(r, T) = 0,$$

where

$$\mathcal{I}(r) = r \quad \text{if } r_l < r < r_u \quad \text{and is zero otherwise.}$$

This is only an approximation to the correct value since in practice the relevant interest rate will have a finite (not infinitesimal) maturity.

18.6 SWAPTIONS, CAPTIONS AND FLOORTIONS

A swaption has a strike rate, r_E , that is the fixed rate that will be swapped against floating if the option is exercised. In a call swaption or **payer swaption** the buyer has the right to become the fixed rate payer; in a put swaption or receiver swaption the buyer has the right to become the payer of the floating leg.

Captions and **floortions** are options on caps and floors respectively. These contracts can be put into the partial differential equation framework with little difficulty. However, these contracts are second order, meaning that their value depends on another, more basic, contract, see Chapter 11. Although the partial differential equation approach is possible, and certainly consistent across instruments, it is likely to be time consuming computationally and prone to serious mispricings because of the high order of the contracts.

18.6.1 Market practice

With some squeezing the Black–Scholes formulæ can be used to value European swaptions. Perhaps this is not entirely consistent, but it is certainly easier than solving a partial differential equation.

The underlying is assumed to be the fixed leg of a par swap with maturity T_S ; call this r_f . It is assumed to follow a lognormal random walk with a measurable volatility. If at time

T the par swap rate is greater than the strike rate r_E then the option is in the money. At this time the value of the swaption is

$$\max(r_f - r_E, 0) \times \text{present value of all future cashflows.}$$

It is important that we are 'modeling' the par rate because the par rate measures the rate at which the present value of the floating legs is equal to the present value of the fixed legs. Thus in this expression we only need worry about the excess of the par rate over the strike rate. This expression looks like a typical call option payoff, all we need to price the swaption in the Black-Scholes framework are the volatility of the par rate, the times to exercise and maturity and the correct discount factors. The payer swaption formula in this framework is

$$\frac{1}{F} e^{-r(T-t)} \left(1 - \left(1 + \frac{1}{2}F \right)^{-2(T_S-T)} \right) (FN(d'_1) - EN(d'_2))$$

and the receiver swaption formula is

$$\frac{1}{F} e^{-r(T-t)} \left(1 - \left(1 + \frac{1}{2}F \right)^{-2(T_S-T)} \right) (EN(-d'_2) - FN(-d'_1))$$

where F is the forward rate of the swap, T_S is the maturity of the swap and

$$d'_1 = \frac{\log(F/E) + \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}} \quad \text{and} \quad d'_2 = d'_1 - \sigma\sqrt{T-t}.$$

These formulae assume that interest payments in the swap are exchanged every six months.

Figure 18.6 shows the Bloomberg swaption valuation page. They use the Black model for pricing.

<HELP> for explanation, <MENU> for similar functions. DL18 M-Mkt OVSW

SWAPTION VALUATION

OPTION TYPE: **E** American/Bermudan/European Swap Curve

SWAP		CURVE # 22
EFFECTIVE DT	9/ 8/00 or 1 Y x 5 Y	B Bid/Ask/Mid
MATURITY	9/ 8/05 NOTIONAL 10000M	CURRENCY: BP BGN CURVE
CALCULATE :1		CURVE DATE: 9/ 8/99
	FIXED FLOATING	Deposit Rates Swap Rates
(1)COUPON	6.80464%	1 W 4.8667 2 Y 6.5800
NOMINAL DATE	9/ 8 + 0.0 bp (2)	1 M 5.0616 3 Y 6.8100
1st COUPON	3/ 8/01	2 M 5.1531 4 Y 6.8100
PAYMENT FREQ	S S	3 M 5.2142 5 Y 6.7300
DAY COUNT	ACT/365	4 M 5.5638 7 Y 6.6100
BUSINESS DAY ADJ.	1	5 M 5.5898 10Y 6.4000
(3)SWAP PREM (PER 100)	0.0000	6 M 5.6250 30Y 5.6600
	OPTION	9 M 5.7589
OPTION TO R P= Pay/R=Receive Fixed		1 Y 6.1200
MODEL 1 1-Black 2-BDT 3-LogNormal 4-NMR		Enter:
SETTLEMENT 9/ 8/99 EXP 9/ 8/00		1 <GO> Update Swap Curve
		2 <GO> View Swap/Option Deltas
		3 <GO> Save OVSW
Volatility 18.8552 Mean Rev Speed N/A		
Opt Prem (per 100) 2.0061036 51.18 bp		
BPV Opt -1884.25 BPV Swap -4058.19		
Delta 0.46 Gamma 7.89 Vega 2.71		

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Princeton: 609-279-3000 Singapore: 226-3000 Sydney: 2-9777-8686 Tokyo: 3-3201-8900 Sao Paulo: 11-3048-4500
1741-53-0 08-Sep-99 18:55:18

Bloomberg

Figure 18.6 Swaption valuation. Source: Bloomberg L.P.

18.7 SPREAD OPTIONS

Spread options have a payoff that depends on the difference between two interest rates. In the simplest case the two rates both come from the same yield curve, more generally the spread could be between rates on different but related curves (yield curve versus swap curve, LIBOR versus Treasury bills), risky and riskless curves or rates in different currencies.

Can we price this contract in the framework we have built up? No. The contract crucially depends on the tilting of the yield curve. In our one-factor world all rates are correlated and there is little room for random behavior in the spread. One way to price such a contract is to use a two-factor interest rate model that captures both the overall rising and falling of the yield curve and also any tilting.

Another method for pricing this contract is to squeeze it into the Black–Scholes-type framework. This amounts to modeling the spread directly as a lognormal (or Normal) variable and choosing suitable rates for discounting. This latter method is the market practice and although intellectually less satisfying it is also less prone to major errors.

18.8 INDEX AMORTIZING RATE SWAPS

A swap is an agreement between two parties to exchange interest payments on some principal, usually one payment is at a fixed rate and the other at a floating rate. In an index amortizing rate (IAR) swap the amount of this principal decreases, or **amortizes**, according to the behavior of an ‘index’ over the life of the swap; typically, that index is a short-term interest rate. The easiest way to understand such a swap is by example, which I keep simple.

Example Suppose that the principal begins at \$10,000,000 with interest payments being at 5% from one party to the other, and $r\%$, the spot interest rate, in the other direction. These payments are to be made quarterly.¹ At each quarter, there is an exchange of $(r - 5)\%$ of the principal. However, at each quarter the principal may also amortize according to the level of the spot rate at that time. In Table 18.1 we see a typical amortizing schedule.

Table 18.1 Typical amortizing schedule.

Spot rate	Principal reduction
less than 3%	100%
4%	60%
5%	40%
6%	20%
8%	10%
over 12%	0%

¹ In which case r would, in practice, be a three-month rate and not the spot rate. The IAR swap is so path dependent that this difference will not be of major importance.

We read this table as follows. First, on a reset date, each quarter, there is an exchange of interest payments on the principal as it then stands. What happens next depends on the level of the spot rate. If the spot interest rate (or whatever index the amortization schedule depends on) is below 3% on the date of the exchange of payments then the principal on which future interest is paid is then amortized 100%; in other words, this new level of the principal is zero and thus no further payments are made. If the spot rate is 4% then the amortization is 60%, i.e. the principal falls to just 40% of its level before this reset date. If the spot rate is 8% then the principal amortizes by just 10%. If the rate is over 12% there is no amortization at all and the principal remains at the same level. For levels of the rate between the values in the first column of the table the amount of amortization is a linear interpolation. This interpolation is shown in Figure 18.7 and the function of r that it represents I call $g(r)$.

So, although the principal begins at \$10,000,000, it can change after each quarter. This feature makes the index amortizing rate swap path dependent.

The party receiving the fixed rate payments will suffer if rates rise because he will pay out a rising floating rate while the principal does not decrease. If rates fall the principal amortizes and so his lower floating rate payments are unfortunately on a lower principal. Again, he suffers. Thus the receiver of the fixed rate wants rates to remain steady and is said to be selling volatility.

In Figure 18.8 is shown the term sheet for a USD IAR swap. In this contract there is an exchange every six months of a fixed rate and six-month LIBOR. This is a vanilla IAR

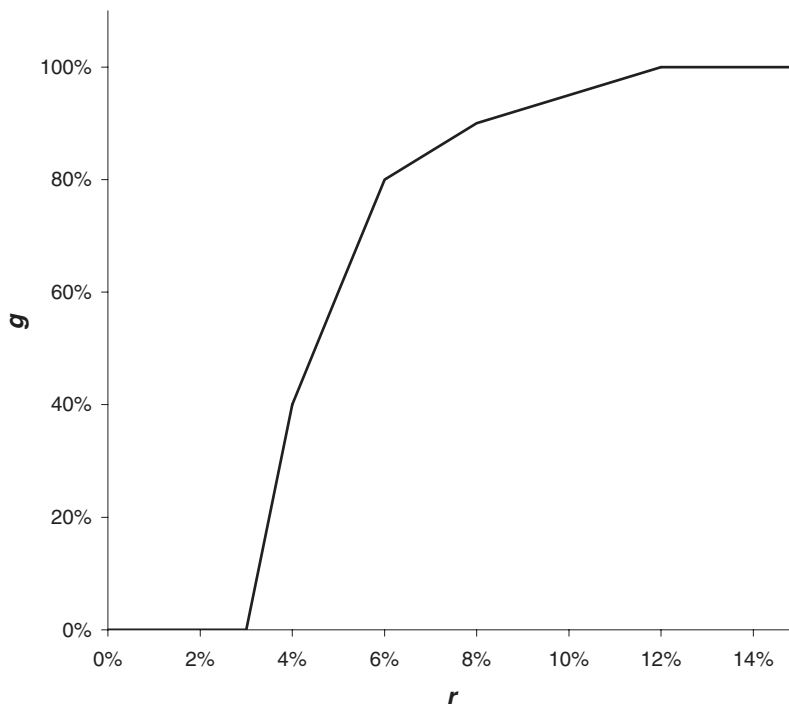
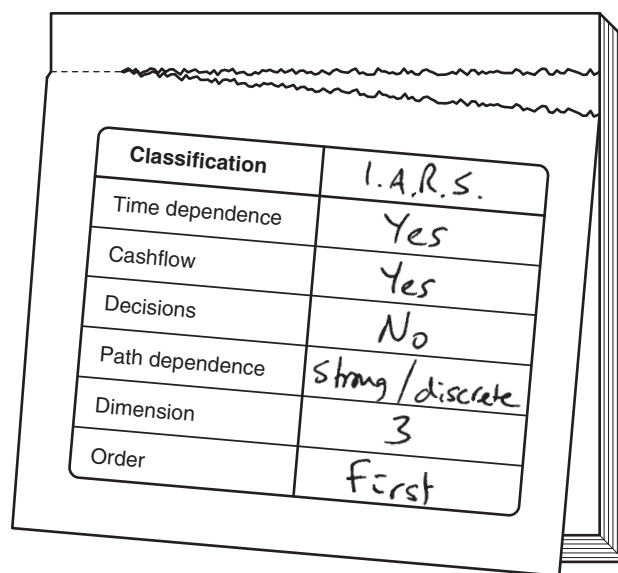


Figure 18.7 A typical amortizing schedule.



Classification	I.A.R.S.
Time dependence	Yes
Cashflow	Yes
Decisions	No
Path dependence	Strong / discrete
Dimension	3
Order	First

swap with no extra features and can be priced in the way described above. Terms in square brackets would be set at the time that the deal was made.



Time Out...

Is there a simple math model?

Not exactly simple, but then not too difficult. This is beyond the scope of this book since it involves path dependency. Let me just say that there is a differential equation formulation, and a tree formulation, and it can also be valued by simulation.

18.8.1 Other features in the index amortizing rate swap

Lockout period There is often a 'lockout' period, usually at the start of the contract's life, during which there are no reductions in the principal. During this period the interest payments are like those of a vanilla swap. Mathematically, we can model this feature by allowing the amortizing schedule, previously $g(r)$, to be time dependent: $g(r, t)$. In this case the amount of amortizing depends on the reset date, t_i , as well as the spot interest rate. Such a model can be used for far more sophisticated structures than the simple lockout period.

*Preliminary and Indicative
For Discussion Purposes Only*

USD Index Amortizing Swap

Counterparties	XXXX The Customer														
Notional Amount	USD 50 millions, subject Amortization Schedule														
Settlement Date	Two days after Trade Date														
Maximum Maturity Date	Five years after Trade Date														
Early Maturity Date	On any Fixing Date leading to a Notional Amount equal to 0														
Payments made by Customer	USD 6m LIBOR paid semiannually, A/360														
Payments made by XXXX	In USD X% p.a. paid semiannually, 30/360														
Index Rate	USD 6m LIBOR														
Base Rate	[]%														
Amortization Schedule (after 1st coupon period)	<table border="1"> <thead> <tr> <th>USD 6m LIBOR – Base Rate</th><th>Amortization</th></tr> </thead> <tbody> <tr> <td>–3%</td><td>–[]%</td></tr> <tr> <td>–2%</td><td>–[]%</td></tr> <tr> <td>–1%</td><td>–[]%</td></tr> <tr> <td>0</td><td>–[]%</td></tr> <tr> <td>1%</td><td>0%</td></tr> <tr> <td>2%</td><td>0%</td></tr> </tbody> </table>	USD 6m LIBOR – Base Rate	Amortization	–3%	–[]%	–2%	–[]%	–1%	–[]%	0	–[]%	1%	0%	2%	0%
USD 6m LIBOR – Base Rate	Amortization														
–3%	–[]%														
–2%	–[]%														
–1%	–[]%														
0	–[]%														
1%	0%														
2%	0%														
Fixing Dates	2 business days before each coupon period														
USD 6m LIBOR	The USD 6m LIBOR rate as seen on Telerate page 3750 at noon, London time, on each Fixing Date														
Documentation	ISDA														
Governing Law	English														

NB If the observed difference falls between two entries of this schedule, the amortization amount is interpolated

This indicative termsheet is neither an offer to buy or sell securities or an OTC derivative product which includes options, swaps, forwards and structured notes having similar features to OTC derivative transactions, nor a solicitation to buy or sell securities or an OTC derivative product. The proposal contained in the foregoing is not a complete description of the terms of a particular transaction and is subject to change without limitation.

Figure 18.8 Term sheet for a USD index amortizing swap.

Clean up Some contracts have that if the principal ever falls to a specified percentage of the original principal then it is reduced all the way to zero.

18.9 **CONTRACTS WITH EMBEDDED DECISIONS**

The following contract is interesting because it requires the holder to make a complex series of decisions on whether to accept or reject cashflows. The contract is path dependent.

This contract, called a **flexiswap**, is a swap with M cashflows of floating minus fixed during its life. The catch is that the holder must choose to accept exactly $m \leq M$ of these cashflows. At each cashflow date the holder must say whether they want the cashflow or not, they cannot later change their mind. When they have taken m of them they can take no more.

Classification	Flexiswap
Time dependence	Yes
Cashflow	Yes
Decisions	Yes
Path dependence	Strong / discrete
Dimension	3 (2 continuous, 1 discrete)
Order	First

18.10 SOME EXAMPLES

The term sheet in Figure 18.9 shows details of a Sterling/Deutschemark deconvergence swap. This contract allows the counterparty to express the view that rates between Germany and the UK will widen. Pricing this contract requires models for both UK and German interest rates and the correlation between them.

*Preliminary and Indicative
For Discussion Purposes Only*

Sterling/Deutschemark Deconvergence Swap

Start Date	10 th March 1999
Maturity Date	10 th March 2003
Counterparty 1	
payments	
Floating rate	3m DEM LIBOR plus SPREAD
SPREAD	2.35%
Counterparty 2	
payments	
Floating rate	3m GBP LIBOR

This indicative termsheet is neither an offer to buy or sell securities or an OTC derivative product which includes options, swaps, forwards and structured notes having similar features to OTC derivative transactions, nor a solicitation to buy or sell securities or an OTC derivative product. The proposal contained in the foregoing is not a complete description of the terms of a particular transaction and is subject to change without limitation.

Figure 18.9 Term sheet for a Sterling/Deutschemark deconvergence swap.

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**USD 1YR Fixed Rate Note with Redemption linked to World
Bank Zero-Coupon Bonds**

The note either redeems at par or the investor is delivered World Bank zero-coupon bonds maturing on either 1st March 2009, 15th April 2009 or 15th July 2009. The choice of redemption is at the Issuer's option. If World Bank bonds are delivered, this will be at effective prices of 57.24% for the March bond, 56.85% for the April bond and 56.05% for the July bond.

Issue Date	15 April 2002
Maturity Date	15 April 2003
Issue price	100%
Coupon	7.00%

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Figure 18.10 Term sheet for a USD fixed rate note.

Figure 18.10 shows a one-year USD fixed rate note with redemption linked to World Bank bonds. The interesting point about this contract is that the issuer of the bond gets to choose whether to redeem at par or to redeem using a choice of three World bank bonds. Obviously the issuer chooses whichever will be cheapest to deliver at the time of redemption. Hence this is an example of a **cheapest-to-deliver** bond.

The term sheet in Figure 18.11 is of a GBP two-year chooser range accrual note linked to GBP LIBOR. The contract pays a daily coupon equivalent to an annual six-month

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GBP 2YR Chooser Range Accrual Note Linked to 6 month GBP LIBOR

The note pays a coupon based on the number of days that 6-month LIBOR sets within an 80 bps range. The range is chosen by the buyer at the beginning of each coupon period.

Issue Date	24 th March 2000
Maturity Date	24 th March 2002
Issue Price	100%
Coupon	$[6 \text{ month LIBOR} + 1.00\%] \times N/D$
N	Number of business days that 6-month LIBOR is within the RANGE
D	Number of business days in the OBSERVATION PERIOD
RANGE	Determined by the buyer two days prior to the beginning of each OBSERVATION PERIOD
OBSERVATION PERIOD	Period 1: 24 th March 2000–24 th September 2000 Period 2: 24 th September 2000–24 th March 2001 Period 3: 24 th March 2001–24 th September 2001 Period 4: 24 th September 2001–24 th March 2002

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Figure 18.11 Term sheet for a chooser range accrual note.

LIBOR + 100 bps. But this is only paid while LIBOR is within an 80 bps range. The novel feature about this range note is that the holder chooses the 80 bps range at the start of each coupon period.

18.11 MORE INTEREST RATE DERIVATIVES...

The following require a stochastic interest rate model for their pricing, they are model dependent.

Accordion swap: A swap (fixed-income instrument, see Chapter 15) whose maturity can be lengthened or shortened at the wish of the holder.

Barrier cap/floor: An interest rate cap or floor with a barrier feature.

Basis swap: A swap in which both legs are floating, of different maturities or currencies, say.

Bermudan swaptions: Bermudan swaptions are like vanilla swaptions in that they give the holder the right to pay (payer swaption) or receive (receiver swaption) the fixed leg of a swap. The Bermudan characteristic allows the holder to exercise into this at specified dates.

Bounded cap or floor: An interest rate cap or floor whose total payout is bounded.

Callable swap: A swap which may be called back by the fixed rate payer.

Constant maturity swap: A swap in which one leg is itself a swap rate of constant tenor (rather than the more standard LIBOR rate).

LIBOR-in-arrears swap: LIBOR-in-arrears swap is an interest rate swap in which the floating payment is made at the same time that it is set. In the plain vanilla swap the rate is fixed prior to the payment, so that a swap with six-monthly payments of six-month LIBOR has the floating rate set six months before it is paid. This subtle difference means that the LIBOR-in-arrears swap cannot be decomposed into bonds and the pricing is not model independent. There will be a slight difference between the vanilla swap and the LIBOR-in-arrears swap. Because the difference depends on the slope of the forward curve, the LIBOR-in-arrears swap is often thought of as a play on the steepening or flattening of the yield curve.

Moving average cap/floor: An interest rate cap/floor with payout determined by an average interest rate over a period.

Puttable swap: A swap which may be called back by the floating rate payer.

Ratchets and one-way floaters: Ratchets and one-way floaters are floating rate notes where the amount of the periodic payments is reset, usually in a monotonically increasing (or decreasing) manner. The amount of the reset will depend on a specified floating interest rate.

Reflex cap/floor: As a cap or floor but with payments depending on a trigger being reached.

Reverse floater: A floating rate note with coupon that rises as the underlying rate falls and vice versa.

Rolling cap/floor: A cap or floor in which the out-of-the-money portion of each payment is carried forward into the next period.

Triggers: Triggers are just like barrier options in that payments are received until (or after) a specified financial asset trades above or below a specified level. For example, the trigger swap is like a plain vanilla swap of fixed and floating until the reference LIBOR rate fixes above/below a specified rate. You can imagine that they come in and out and up and down varieties.

18.12 SUMMARY

There are a vast number of contracts in the fixed-income world. It is an impossible task to describe and model any but a small quantity of these. In this chapter I have tried to show two of the possible approaches to the modeling in a few special cases. These two approaches to the modeling are the consistent way via a partial differential equation or the practitioner way via the Black–Scholes equity model and formulæ. The former is nice because it can be made consistent across all instruments, but is dangerous to use for liquid and high-order contracts. Save this technique for the more complex, illiquid and path-dependent contracts. The alternative approach is, as everyone admits, a fudge, requiring a contract to be squeezed and bent until it looks like a call or a put on something vaguely lognormal. Although completely inconsistent across instruments it is far less likely to lead to serious mispricings.

The reader is encouraged to find out more about the pricing of products in these two distinct ways. Better still, the reader should model new contracts for himself as well.

FURTHER READING

- Black (1976) models the value of bond options assuming the bond is a lognormal asset.
- See Hull & White (1996) for more examples of pricing index amortizing rate swaps.
- Everything by Jamshidian on the pricing of interest rate derivatives is popular with practitioners. See the bibliography for some references.
- The best technical book on interest rate derivatives, their pricing and hedging, is by Rebonato (1998).
- See <http://my.dreamwiz.com/stoneq/products> for a comprehensive list of interest rate derivatives.
- Hagan (2002) explains how to express the volatility risk of exotics in terms of their natural vanilla hedging instruments.

- For an explanation of convexity corrections for several instruments see Hagan (2003).
- See Jäckel (2003) for cap pricing.
- See Berrahoui (2004) for CMS pricing.
- Jäckel & Kawai (2005) present formulæ for the prices of interest rate futures contracts allowing for volatility skew.

EXERCISES

1. Write down the problem we must solve in order to value a puttable bond.
2. Derive a relationship between a floorlet and a call option on a zero-coupon bond.
3. How would a collar be valued practically? What is the explicit solution for a single payment?
4. When an index amortizing rate swap has a lockout period for the first year, we must solve

$$\frac{\partial V}{\partial t} + \frac{1}{2}w^2 \frac{\partial^2 V}{\partial r^2} + (u - \lambda w) \frac{\partial V}{\partial r} - rV = 0,$$

with jump condition

$$V(r, P, t_i^-) = V(r, g(r, i)P, t_i^+) + (r - r_f)P,$$

where

$$g(r, i) = 1 \quad \text{if } t_i < 1,$$

and with final condition

$$V(r, P, T) = (r - r_f)P.$$

In this case, reduce the order of the problem using a similarity reduction of the form

$$V(r, P, t) = PH(r, t).$$

5. Find the approximate value of a cashflow for a floorlet on the one-month LIBOR, when we use the Vasicek model.